SmartBin

RESPONSIBILITIES	COLLABORATORS
- Store bin location and zone - Track current fill level (%) - Maintain bin status (Active/Full/Offline) - Provide bin ID and metadata	- IoTSensor - FillLevelRecord - Alert - Dashboard

IoTSensor

RESPONSIBILITIES	COLLABORATORS
- Measure bin fill level via ultrasonic sensing - Transmit data wirelessly to CentralServer - Report its own health/status - Flag itself as malfunctioning if invalid	- SmartBin - CentralServer - FillLevelRecord - MaintenanceReport

FillLevelRecord

RESPONSIBILITIES	COLLABORATORS
- Log fill-level reading with timestamp - Store binID and sensorID reference - Provide historical data for analytics - Archive or correct old readings	- IoTSensor - CentralServer - Dashboard - Alert

CentralServer

RESPONSIBILITIES	COLLABORATORS
- Receive and validate incoming sensor data - Log data to FillLevelRecord - Compare fill level to threshold (80%/100%) - Trigger alert generation - Notify Dashboard and RouteOptimizer	- IoTSensor - FillLevelRecord - Alert - Dashboard - RouteOptimizer

Alert

RESPONSIBILITIES	COLLABORATORS
- Store alert type (Warning/Critical/Malfunction) - Record timestamp and binID - Track alert status (Active/Resolved) - Support acknowledgement and resolution	- CentralServer - Dashboard - MaintenanceReport - RouteOptimizer

MaintenanceReport

RESPONSIBILITIES	COLLABORATORS
- Create maintenance ticket for faulty sensor - Track repair status (Open/In-Progress/Closed) - Link to Technician and Sensor - Log repair history	- IoTSensor - Alert - Dashboard - MaintenanceTechnician

Dashboard

RESPONSIBILITIES	COLLABORATORS
- Display real-time bin fill map - Show active alerts to supervisor - Mark sensors offline when flagged - Refresh status after each data cycle	- CentralServer - Alert - SmartBin - CitySupervisor (actor)

RouteOptimizer

RESPONSIBILITIES	COLLABORATORS
- Receive critical alert trigger from CentralServer - Calculate optimal truck route for flagged bins - Dispatch route to Fleet Management - Update route when new critical alerts arise	- CentralServer - Alert - SmartBin - FleetManager (external)